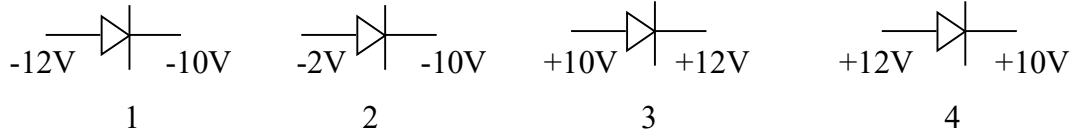


1. The average of the function $f(x) = \sin x$ in the interval $(0, \pi)$ is:
 A) 0 B) $\frac{2}{\pi}$ C) $\frac{1}{2}$ D) $\frac{1}{\pi}$
2. The modulus of complex number $\frac{(2+i)^2}{(3-i)^2}$ is:
 A) $\sqrt{2}$ B) $\sqrt{\frac{3}{2}}$ C) $\frac{3}{2}$ D) $\frac{1}{2}$
3. The residue of $\frac{1}{\tan z}$ at $z = 0$ is:
 A) 1 B) -1 C) π D) $\frac{\pi}{2}$
4. Find $\text{grad}(\phi^n)$, if $\phi = \log r$:
 A) $\frac{\vec{r}}{r}$ B) $\frac{\vec{r}}{r^2}$ C) $\frac{\vec{r}}{r^3}$ D) 0
5. If $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$ then angle between the vectors $\vec{A}$ and $\vec{B}$ is:
 A) 0 B) 90 C) 180 D) None of these
6. If $J_n(x)$ represents Bessel's function, $J_{1/2}(\pi/2)$ is:
 A) $\frac{2}{\pi}$ B) 1 C) -1 D) $\frac{\pi}{2}$
7. Discharge of a charged capacitor along an inductor with finite resistance is identical to the ----harmonic oscillator.
 A) Simple B) Damped C) Forced D) None of these
8. Differential equation in terms of radial and angular coordinates for the motion of a particle of mass 'm' in a central force field is ... (L is angular momentum)
 A) $\frac{d^2}{d\theta^2}\left(\frac{1}{r}\right) + \frac{1}{r} - \frac{mr^2}{L^2}F(r) = 0$ B) $\frac{d^2}{d\theta^2}\left(\frac{1}{r}\right) + \frac{1}{r} + \frac{m}{L^2 r^2}F(r) = 0$
 C) $\frac{d^2}{d\theta^2}(r) + r - \frac{mr^2}{L^2}F(r) = 0$ D) $\frac{d^2}{d\theta^2}\left(\frac{1}{r}\right) + \frac{1}{r} + \frac{mr^2}{L^2}F(r) = 0$
9. If A and B represent the operators of two dynamic variables and they commute, then the value of poisson bracket of A and B is:
 A) $i\hbar$ B) $-i\hbar$ C) 0 D) 1

10. Canonical momenta for a charged particle in an electromagnetic field is:
- A) $\frac{1}{2}mv^2 + \frac{qv}{c^2}A$ B) $mv + \frac{q}{c}A$
- C) $mv - \frac{q}{c}A$ D) $\frac{1}{2}mv^2 - \frac{qv}{c^2}A$
11. The quantity that remains invariant under Galilean transformation is:
- A) velocity B) momentum C) Force D) All of these
12. The dominant mode of propagation along a circular wave guide is:
- A) TM_{01} B) TE_{01} C) TM_{11} D) TE_{11}
13. The fundamental phenomenon that is utilized in a optical wave guide is---- of light.
- A) Interference B) Total internal reflection
- C) Diffraction D) Doppler effect
14. The magnitude of vector potential at a point (interior) with a distance r away from the axis of an infinite solenoid with n turns per unit length and radius R carrying a current I
- A) $\frac{\mu_0 n I r}{2}$ B) $\frac{\mu_0 n I R^2}{2r}$ C) $\frac{\mu_0 n I r^2}{2R}$ D) $\frac{\mu_0 n I R}{2}$
15. The energy transported by an electromagnetic wave per unit time through unit area is represented as ---- vector.
- A) Poynting B) Polarisation vector
- C) Magnetisation vector D) Displacement vector
16. If magnetic monopole is found to existing, then which of the Maxwell equation has to be modified?
- A) $\nabla \cdot D = \rho$ B) $\nabla \cdot B = 0$
- C) $\nabla \times E = -\frac{\partial B}{\partial t}$ D) $\nabla \times H = J + \frac{\partial D}{\partial t}$
17. For reflection and refraction of light at a dielectric interface:
- A) Tangential components of D and B are continuous
- B) Tangential components of D and normal components of B are continuous
- C) Normal components of D and B are continuous
- D) Normal components of D and tangential components of B are continuous
18. An electric dipole of charge 'q' and length 'a' is enclosed in a closed cavity. Then the net electric flux passing out of the cavity is :
- A) $\frac{q}{\epsilon_0}$ B) $\frac{2\pi r^2 q}{\epsilon_0}$ C) $\frac{4\pi r^2 q}{\epsilon_0}$ D) Zero
19. The principal outcome of Michelson's Morley experiment is that it established the :
- A) Wave nature of light B) Absence of hypothetical medium ether
- C) Uncertainty principle D) Wave nature of particles

20. The electric potential due to an electric dipole at an equatorial point which is at a very large distance r much greater than the dipole length is proportional to:
 A) r B) r^{-1} C) r^{-2} D) r^{-3}
21. Magnetic induction $\vec{B}$ associated with a magnetic vector potential $\vec{A} = 2x\hat{i} - y\hat{j} + z\hat{k}$ is :
 A) $2\hat{i} - \hat{j} + \hat{k}$ B) $2\hat{i} + \hat{j} + \hat{k}$ C) zero D) $-2\hat{i} + \hat{j} - \hat{k}$
22. The existence of zero point energy for a linear harmonic oscillator is a consequence of:
 A) Pauli exclusion principle B) Special theory of relativity
 C) Matter waves D) Uncertainty principle
23. The explicit expression for the operator $\left(\frac{d}{dx} + x\right)^2$ is given by:
 A) $\frac{d^2}{dx^2} + 2x\frac{d}{dx} + x^2$ B) $\frac{d^2}{dx^2} + x^2$
 C) $\frac{d^2}{dx^2} + 2x\frac{d}{dx} + x^2 + 1$ D) $\frac{d^2}{dx^2} + 2x\frac{d}{dx} + x^2 - 1$
24. The value of $[L_x, r^2]$ is:
 A) $i\hbar y$ B) $i\hbar p_y$ C) $-i\hbar x$ D) 0
25. In partial wave analysis , scattering amplitude may be written as :
 A) $\frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) P_1(\cos\theta) e^{i\delta_l} \sin\delta_l$
 B) $\frac{1}{k^2} \sum_{l=0}^{\infty} (2l+1) P_1(\cos\theta) e^{i\delta_l} \sin\delta_l$
 C) $\frac{1}{k^2} \sum_{l=0}^{\infty} (2l+1) P_1^2(\cos\theta) e^{i\delta_l} \sin\delta_l$
 D) $\frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) P_1^2(\cos\theta) e^{i\delta_l} \sin\delta_l$
26. Multiplicity of terms is defined as:
 A) $r = 2S$ B) $r = 2S - 1$ C) $r = 2S + 1$ D) $r = S + 1$
27. The only resonance line of sodium is:
 A) $4p \rightarrow 3d$ B) $3s \rightarrow 3p$ C) $4p \rightarrow 4s$ D) $4p \rightarrow 3s$
28. The Lande g factor for the level 3D_3 is :
 A) $\frac{2}{3}$ B) $\frac{3}{2}$ C) $\frac{3}{4}$ D) $\frac{4}{3}$
29. The degeneracy of spectral term 3F is:
 A) 7 B) 9 C) 15 D) 21

30. The normal Zeeman effect is:
 A) Observed only in atoms with an even number of electrons
 B) Observed only in atoms with an odd number of electrons
 C) A confirmation of space quantization
 D) A confirmation of spin quantization
31. Which of the following molecules does **not** exhibit a rotational spectrum?
 A) H_2 B) CO C) HCl D) HBr
32. The effect used to study the energy levels of homonuclear molecules is---- effect.
 A) Stark B) Zeeman
 C) Pachen Back D) Raman
33. In rotation-vibration (IR) spectra of diatomic molecules the lines corresponding to $\nabla J = -1$ is called:
 A) R branch B) Q branch C) P branch D) None of these
34. NMR spectrum is obtain in ---- region.
 A) Microwave B) Radiowave C) Infrared D) UV-visible
35. The selection rule for ESR spectroscopy is:
 A) $\nabla m_s \neq 0$ B) $\nabla m_l = \pm 1$ C) $\nabla m_l = \pm 0$ D) $\nabla m_s = \pm 1$
36. In Debye's theory of specific heat of solids, the atomic oscillators obey:
 A) Fermi-Dirac statistics
 B) Bose-Einstein statistics
 C) Maxwell Boltzmann statistics
 D) All the above
37. The quantum of thermal energy is termed as:
 A) phonon B) photon C) graviton D) pulser
38. The number Bravais lattices that can exist in nature is:
 A) 7 B) 14 C) 32 D) 27
39. The susceptibility of a superconductor is:
 A) Positive and small B) Positive and unity
 C) Negative and small D) Negative and unity
40. The reciprocal lattice to an fcc lattice is:
 A) fcc B) bcc C) SC lattice D) None of these
41. Due to Frenkel defect the density of the ionic solids:
 A) Increases B) Decreases
 C) Does not change D) May increase or decrease

42. Semi-empirical mass formula is known as:
 A) Weizsaecker formula B) Frank-Condon formula
 C) Baldwin formula D) Gomows relation
43. The packing fraction is zero for:
 A) ${}^4_2\text{He}$ B) ${}^{16}_8\text{O}$ C) ${}^{12}_6\text{C}$ D) ${}^{14}_7\text{N}$
44. Volume of nucleus in an atom is proportional to:
 A) Atomic number B) Mass number
 C) Neutron number D) Difference between atomic number and neutron number
45. The binding energy per nucleon is maximum for the nucleus:
 A) ${}^{56}_{26}\text{Fe}$ B) ${}^4_2\text{He}$ C) ${}^{208}_{82}\text{Pb}$ D) ${}^{101}_{42}\text{Mo}$
46. The reaction $e^+ + e^- \rightarrow \mu^+ + \pi^-$ is forbidden because of the law of:
 A) Baryon number conservation
 B) Momentum conservation
 C) Muon number conservation
 D) None of the above
47. A p type semiconductor is electrically:
 A) positive B) negative C) neutral D) unstable
48. Which of the following circuits are forward biased?

 1 2 3 4
- A) 1 & 4 only B) 3 & 4 only C) 2 & 4 only D) 2 & 3 only
49. The type of diodes suitable in optical communication systems for handling large power is ---- diodes.
 A) PIN B) Zener C) Avalanche D) None of these
50. How many flip-flops are required to build a binary counter circuit to count from 0 to 1023?
 A) 1 B) 6 C) 10 D) 24
51. In a microprocessor the register which holds the address of the next instruction to be fetched is:
 A) Accumulator B) Program counter
 C) Stack pointer D) Instruction register

52. 8085 is a ----bit microprocessor.
 A) 4 bit B) 8 bit C) 16 bit D) 32 bit
53. Which of the following are universal gate?
 1. NOT 2. NAND 3. XOR 4. NOR
 A) 1 & 2 only B) 2, 3 & 4 only C) 3 & 4 only D) 2 & 4 only
54. In an op-amp integrator circuit the feedback element is:
 A) Resistor B) Capacitor C) Diode D) Inductor
55. An n-channel D-MOSFET with a positive V_{GS} is operating in:
 A) Depletion mode B) Enhancement mode
 C) Cut off D) Saturation
56. If the voltage gain doubles, the decibel voltage gain increases by:
 A) a factor of 2 B) 3 dB C) 6 dB D) 10 dB
57. A voltage follower has a voltage gain:
 A) Much less than one B) 1
 C) Greater than one D) Infinite
58. A D/A converter with four inputs have ---- output values.
 A) Two B) Four C) Eight D) Sixteen
59. When a large sine wave drives a Schmitt trigger, the output is a
 A) Rectangular wave B) Triangular wave
 C) Rectified sine wave D) Series of ramp
60. Given $z = 6e^{\pi i/3}$. Evaluate $|e^{iz}|$.
 A) $e^{-3\sqrt{3}}$ B) $e^{3(i-\sqrt{3})}$ C) $e^{3i+3\sqrt{3}}$ D) $e^{+3\sqrt{3}}$
61. Evaluate $L(4 \cos^2 2t)$. [Hint: L represents Laplace transform].
 A) $\frac{2}{s} + \frac{s}{s+4}$ B) $\frac{2}{s} + \frac{2s}{s^2+16}$ C) $\frac{2}{s} - \frac{s}{s^2+4}$ D) $\frac{2}{s} + \frac{2s^2}{s^2+16}$
62. If $f(z) = u + iv$ is analytic in a region R, then u and v are harmonic in R. This is true for---- theorem.
 A) Morera's B) Parseval's
 C) Cauchy Riemann D) Liouville's
63. Find the residue of $f(z) = \frac{z^2}{(z-1)(z-2)(z-3)}$ at the poles.
 A) 1, 4, 3/2 B) -1, -2, -3 C) 1, 2, 3 D) 1/2, -4, 9/2

64. Evaluate $\int \left(\vec{r} \times \frac{d^2 \vec{r}}{dt^2} \right) dt$
- A) $\vec{r}$ B) $\vec{r} \times \frac{d\vec{r}}{dt}$
 C) $\vec{r}^2 \times \frac{d\vec{r}}{dt}$ D) Cannot be evaluated
65. Match the following :
- | | |
|-----------------------|--|
| List I | List II |
| a. 1D heat equation | 1. $\nabla^2 \phi = -\frac{\rho}{\epsilon_0}$ |
| b. 2D heat equation | 2. $\frac{\partial u}{\partial t} = \alpha \nabla^2 u$ |
| c. Laplace equation | 3. $\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$ |
| d. Poisson's equation | 4. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ |
- A) a-2, b-1, c-3, d-4 B) a-3, b-4, c-2, d-1
 C) a-2, b-3, c-1, d-4 D) a-3, b-2, c-4, d-1
66. In a simple harmonic oscillator, the phase difference between displacement and acceleration is ----.
- A) 0 B) $\pi/2$ C) π D) 2π
67. A particle moving in a force field has the equation of its orbit $r = a\theta$. Find the law of force. (Hint. Use equation of orbit with angular momentum per unit reduced mass)
- A) $f(r) = -\frac{l^2}{\mu} \left(\frac{2a}{r^5} + \frac{1}{r^3} \right)$ B) $f(r) = -\frac{l^2}{\mu} \left(\frac{2}{r^5} + \frac{1}{r^3} \right)$
 C) $f(r) = -\frac{l^2}{\mu} \left(\frac{2a^2}{r^5} + \frac{1}{r^3} \right)$ D) $f(r) = \frac{l}{\mu} \left(\frac{2a^2}{r^5} + \frac{1}{r^3} \right)$
68. A particle constrained to move on the inner surface of a parabolic of revolution has ---- degrees of freedom.
- A) 1 B) 2 C) 3 D) 4
69. Which is the conserved quantity for the below given Lagrangian of a free particle in spherical polar coordinate? $L = \frac{1}{2} m (\dot{r}^2 + r\dot{\theta}^2 + r^2\dot{\phi}^2 \sin^2 \theta)$
- A) $\frac{\partial L}{\partial r}$ B) $\frac{\partial L}{\partial \theta}$ C) $\frac{\partial L}{\partial \phi}$ D) $\frac{\partial L}{\partial \phi} + \dot{r}\dot{\theta}$

70. A beam of current I of relativistic particles of charge e and rest mass m_0 hits an absorbing target. Each particle has a kinetic energy E_k . Obtain the pressure on the target:
- A) $\frac{I}{ec} \sqrt{E_k(E_k + 2m_0c^2)}$ B) $Iec \sqrt{E_k(E_k + 2m_0c^2)}$
- C) $\frac{I}{ec} \sqrt{(E_k^2 + (2m_0c^2)^2)}$ D) $\frac{I}{ec} E_k$
71. Choose the first order correction to the energy in non-degenerate time-independent perturbation theory:
- A) $E_n^{(1)} = \langle n^{(0)} | V | n^{(0)} \rangle$ B) $E_n^{(1)} = 0$
- C) $E_n^{(1)} = 1$ D) $E_n^{(1)} = \langle n^{(1)} | V | n^{(1)} \rangle$
72. If there are two different wave functions corresponding to the same energy, then any linear combination of them will also be a valid wave function for the same energy. The system is said to be----?
- A) Non-degenerate B) Degenerate
- C) Orthogonal D) None of these
73. Calculate the lowest possible total energy consistent with quantum rules of a collection of 7 non interacting electrons of mass m , and spin $\frac{1}{2}$ when placed in a 1D infinite square well potential of width L .
- A) $\frac{30 \hbar^2}{8mL^2}$ B) $\frac{11 \hbar^2}{2mL^2}$ C) $\frac{7 \pi^2 \hbar^2}{8mL^2}$ D) $\frac{11 \pi^2 \hbar^2}{2mL^2}$
74. The position operator in momentum space representation is:
- A) $-i\hbar \frac{\partial}{\partial p}$ B) $\hbar \frac{\partial}{\partial p}$ C) $i\hbar \frac{\partial}{\partial p}$ D) $-i\hbar \frac{\partial}{\partial x}$
75. The expectation values of the dot product of the spin operators of the two electrons in a Helium atom for the singlet and triplet states respectively are:
- A) $\frac{-3}{4} \hbar^2 ; \frac{1}{4} \hbar^2$ B) $\frac{-3}{2} \hbar^2 ; \frac{1}{4} \hbar^2$
- C) $\frac{3}{4} \hbar^2 ; \frac{1}{2} \hbar^2$ D) $\frac{-1}{4} \hbar^2 ; \frac{1}{4} \hbar^2$
76. The Hamiltonian for a spin $\frac{1}{2}$ particle at rest is given by $H = E_0(\sigma_z + \alpha \sigma_x)$, E_0 and α are constants and σ_z, σ_x are Pauli spin matrices. Find the eigen values:
- A) $(1 - \alpha)$ and $(1 + 2\alpha)$ B) E_0 doubly degenerate
- C) $\pm E_0 \sqrt{1 + \alpha^2}$ D) $E_0(1 \pm \frac{1}{2} \alpha^2)$
77. An anisotropic harmonic oscillator described by the potential $V(x, y, z) = \frac{1}{2} m \omega^2 x^2 + 2m \omega^2 y^2 + 8m \omega^2 z x^2$. Calculate the ground state energy (in units of $\hbar \omega$).
- A) $3/2$ B) $1/2$ C) $7/2$ D) $5/2$

78. Which of the commutation relation is **incorrect**?
 A) $[a, a^\dagger] = 1$ B) $[a^\dagger a, a] = -a$ C) $[a^\dagger a, a^\dagger] = -a^\dagger$ D) $[a^\dagger, a] = -1$
79. A particle scatters off a repulsive spherical potential $V(r)$. Calculate the ratio of total scattering cross section (σ) to classical scattering cross section of a sphere of radius 'a' when $V_0 \rightarrow \infty$. Give below:

$$V(r) = \begin{cases} V_0 & \text{for } r < a \\ 0 & \text{for } r \geq a \end{cases}, \text{ where } V_0 \text{ and } a \text{ are positive constants for}$$

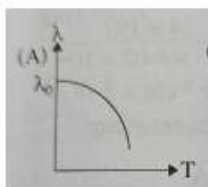
$$\sigma = 4\pi a^2 \left(\frac{1}{ka} \tan h ka - 1 \right)^2, \text{ in the low energy limit}$$

$$k^2 = \frac{2m}{\hbar^2} (V_0 - E), \text{ where } k \text{ is the wave number and } E \text{ is the energy of the particle:}$$
 A) 3 B) π C) 1 D) 4
80. The normalized ground state wavefunction of H-atom is $\psi(r) = \frac{1}{\sqrt{4\pi}} \frac{2}{a^{3/2}} e^{-r/a}$, where a is Bohr radius and r is the distance of the electron from the nucleus located at the origin. Find the expectation value $\langle \frac{1}{r^2} \rangle$:
 A) $\frac{8\pi}{a^2}$ B) $\frac{2}{a^2}$ C) $\frac{2\pi}{a^2}$ D) $\frac{4}{a^2}$
81. A particle of mass m is confined in the ground state of a 1D box extending from $x = -2L$ to $x = +2L$, wherein its wavefunction is $\psi(r) = \psi_0 \cos \frac{\pi x}{4L}$ where ψ_0 is a constant. Find the energy eigen value corresponding to this state.
 A) $\frac{\pi^2 \hbar^2}{32 mL^2}$ B) $\frac{\pi^2 \hbar^2}{128 mL^2}$ C) $\frac{n^2 \hbar^2}{8 mL^2}$ D) $\frac{\hbar^2}{8 mL^2}$
82. In which wavelength region the Planck's law reduces to Wien's law when applied to solar radiation?
 A) Microwave B) Visible C) Infrared D) Ultraviolet
83. Which is true as far as temperature and entropy of photon gas is considered?
 A) Proportional to T B) Proportional to T^2
 C) Proportional to T^3 D) Not known
84. Find the free energy of N independent oscillations of a harmonic oscillator. (The partition function = $kT/\hbar \omega$, where the symbols have their usual meaning).
 A) $3/2 NkT$ B) $kT \ln(kT/\hbar \omega)$
 C) $(kT/\hbar \omega)^{3N}$ D) $-NkT \ln(kT/\hbar \omega)$
85. Find the ratio of magnetic induction at the centre of a very long solenoid to that at the extreme edge:
 A) 4:1 B) 1:2 π C) 2:1 D) 1:4

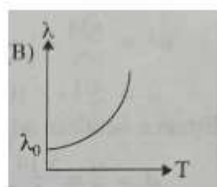
86. The amount of scattered light in Rayleigh scattering is proportional to:
 A) $\frac{1}{\lambda^4}$ B) $\frac{1}{\lambda^3}$ C) $\frac{1}{\lambda^2}$ D) $\frac{1}{\lambda}$
87. A point charge is kept at the midpoint between two large parallel grounded conducting plates. Assuming no gravity, the charge is displaced a little towards the right plate. The charge will now ----?
 A) Move towards the left plate
 B) Accelerate towards the right plate
 C) Stay where it is
 D) Oscillate between the plates
88. Obtain the charge density due to an electric potential $\phi = \phi_0(x^2 + y^2 + z^2)$, ϕ_0 is a constant:
 A) $-2\epsilon_0\phi_0$ B) $-6\phi_0$ C) $-6\epsilon_0\phi_0$ D) Zero
89. If a proton is moved against the Coulomb force of an electric field, then ----?
 A) Work is done by the electric field
 B) Energy is used from some outside source
 C) The strength of the field is decreased
 D) No effect is observed
90. A perfectly conducting spherical shell rotating about Z-axis carries a surface charge density $\sigma(\theta) = \sigma_0 \cos \theta$. Choose the correct option about the electromagnetic momentum:
 A) Is zero because fields inside a conductor vanish.
 B) Is zero because the net charge on the sphere is zero.
 C) Is non-zero because rotating charge produces both E and B, whose cross product stores momentum.
 D) Is non-zero because magnetic dipole moment and electric dipole moment are perpendicular
91. Consider an accelerating point charge. Arrange the following fields in the increasing order of their distance dependence (Hint: least steep fall-off first).
 1. Radiation field 2. Velocity (induction) field 3. Coulomb near field
 A) $2 \rightarrow 1 \rightarrow 3$ B) $3 \rightarrow 2 \rightarrow 1$ C) $1 \rightarrow 3 \rightarrow 2$ D) $1 \rightarrow 2 \rightarrow 3$
92. A plane electromagnetic wave $E = 100 \cos(6 \times 10^8 t + 4x)$ V/m propagates in a medium. Estimate the dielectric constant.
 A) 1.54 B) 2.0 C) 4.0 D) 1.33
93. In He-Ne laser, the laser transition takes place in:
 A) First in Ne, then in He B) Ne only
 C) He only D) First in He, then in Ne

94. Estimate the approximate number of photons emitted per second from a 1W Ar-ion laser operating at 488 nm.
 A) 10.2×10^{19} B) 10.2×10^{17} C) 2.5×10^{18} D) None of these
95. The tensor component of nuclear force may be inferred from the fact that, deuteron nucleus (${}_1\text{H}^2$) :
 A) Has only two nuclear bound state
 B) Has one bound state with total spin 1
 C) Has non-zero ground state electric quadrupole moment
 D) None of these
96. Choose the correct option for a stationary particle in free space when decayed into two photons:
 A) It is a charged particle
 B) Particle has half integral spin
 C) Particle obeys Bose-Einstein statistics
 D) None of these
97. When a nuclear reactor starts up, which of the following is made slightly greater than 1, and then gradually decreases to 1 for a steady chain reaction?
 A) Fission cross section B) Neutron reproduction factor
 C) Moderator speed D) Combustion factor
98. Number of flavours of in connection with the existence of quarks is:
 A) 6 B) 4 C) 2 D) 16
99. The mean momentum of a nucleon in a nucleus with mass number A varies as:
 A) A^2 B) $A^{-1/3}$ C) $A^{1/3}$ D) $A^{-2/3}$
100. Calculate the distance of closest approach when protons of energy 1MeV undergo Rutherford scattering by a nucleus ($Z=79$).
 A) 11.4 nm B) 1138 Å C) 11.4 pm D) 113.8 fm
101. The Baryon number of electron, the lepton number of proton, the lepton number of electron and the baryon number of proton are respectively:
 A) Zero, zero, one, one B) One, zero, one, zero
 C) One, zero, zero, one D) One, one, zero, one
102. The C-14 nuclide in an antique wooden decorative piece is measured to have a specific activity of $3/5$ of that in freshly fallen trees. Half life of C-14 is 3010 years. How old the wooden item is ---- years.
 A) 4018 B) 2220 C) 4.1×10^2 D) 5017

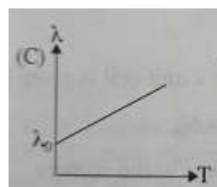
103. An unstable particle of rest energy 1000 MeV decays into a μ -meson and a neutrino with a mean lifetime of $0.01 \mu\text{s}$. Calculate the approximate mean decay distance when the particle has a momentum of $1000/c \text{ MeV}$:
 A) 6 m B) 9 m C) 3m D) 2 m
104. Energy released in the explosion of an atom bomb consisting of ^{235}U is 10^{14}J . Each ^{235}U releases 3 neutrons per fission along with 200MeV . Calculate the total number of neutrons released in this explosion:
 A) 1.7×10^{24} B) 9.4×10^{24} C) 3.7×10^{24} D) 7.9×10^{24}
105. The basis vectors of a 2D crystal lattice are $\vec{a}=2\hat{x}$ and $\vec{b}=\hat{x}+2\hat{y}$. Get the basis vectors in the reciprocal lattice:
 A) $(2\hat{x} - \hat{y}), \hat{y}$ B) $\frac{\pi}{2}(\hat{x} + 2\hat{y}), \pi\hat{y}$
 C) $\frac{\pi}{2}(2\hat{x} - 2\hat{y}), \pi\hat{y}$ D) $\frac{\pi}{2}(2\hat{x} - \hat{y}), \pi\hat{y}$
106. The absolute value of velocity of electron corresponding to the point of inflexion at E-k diagram is:
 A) Minimum B) Zero C) Maximum D) Not known
107. Match the following:
 List I (crystal structures) List II (coordination number)
 a. simple cube 1. 12
 b. body centered cube 2. 6
 c. face centered cube 3. 4
 d. diamond 4. 8
 A) a-2, b-1, c-4, d-3 B) a-4, b-3, c-2, d-1
 C) a-2, b-4, c-1, d-3 D) a-1, b-4, c-2, d-3
108. The electron density in metal A is twice that in metal B at absolute zero. Compare their Fermi energies:
 A) $2^{2/3}$ B) $2^{3/2}$ C) $1/2^{3/2}$ D) 2
109. Identify the correct London penetration depth-Temperature graph:



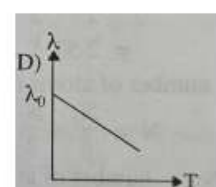
A)



B)



C)



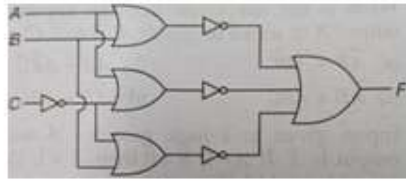
D)

110. The energy dispersion relation for a certain electronic band near a critical point is modelled as a function of crystal momentum magnitude k by the expression $E(k) = ak^2 - bk^4$, where a and b are positive constants. What is the maximum energy of the electrons and at what value of k does this maximum occur?
- A) $\frac{a^2}{4b}, \sqrt{\frac{a}{2b}}$ B) $\frac{a^2}{b}, \sqrt{\frac{a}{2b}}$ C) $\frac{a^2}{8b}, \sqrt{\frac{a}{b}}$ D) $\frac{a}{4b}, \sqrt{\frac{a}{2b}}$
111. In a powder X-ray diffraction pattern of lead using X-rays of wavelength $1.54 \text{ \AA}$, the (220) Bragg reflection is observed at a diffraction angle of 30° . Using this information, calculate the atomic radius of lead:
- A) $3.13 \text{ \AA}$ B) $5.41 \text{ \AA}$ C) $1.54 \text{ \AA}$ D) Data inadequate
112. The potential energy representing the interaction between two atoms is $U(r) = -\frac{2}{r^4} + \frac{3}{r^6}$. Calculate the equilibrium distance between them and the potential energy in equilibrium:
- A) $\frac{3}{2}$ and $\left(\frac{-2}{3}\right)^5$ B) $\sqrt{\frac{3}{2}}$ and $\left(\frac{2}{3}\right)^4$ C) $3/2$ and $(-2/3)$ D) $\frac{2}{3}$ and $\left(\frac{-2}{3}\right)^5$
113. Assertion (A): If a sample of Silicon of thickness $200\mu\text{m}$ is doped with 10^{23} Phosphorous atoms/ m^3 and when kept in a magnetic field of 0.2 T under the passage of a current of 1mA , the Hall voltage produced is 6.25 mV .
- Reason (R): $V_H = \frac{BI}{\text{net}}$
- A) Both A and R are true and R is the correct explanation of A
 B) Both A and R are true, but R is not the correct explanation of A
 C) A is true but R is false
 D) A is false but R is true
114. Identify the logic gate corresponding to the given truth table:

Input		Output
0	0	1
0	1	0
1	0	0
1	1	1

- A) XOR B) RS flip flop
 C) Mod-3 Counter D) XNOR

115. Choose the **incorrect** logic expression for the given logic diagram:



- A) $F = (A+B)' + (A+C')' + (B+C')'$
 B) $F = (A'B') + C (A'+B')$
 C) $F = (A'B') + (A'C) + (CB')$
 D) $F = A + B + C'$
116. Slew rate of an Op-Amp is $2.2V/\mu s$. If the peak output is 14V, calculate the power band width:
 A) 0.06 MHz B) 6 MHz C) 0.03 MHz D) 0.3 MHz
117. The data bus width of 204 x 8 bits is:
 A) 8 B) 12 C) 20 D) 16
118. Calculate the dynamic resistance of an ideal p-n junction diode with a forward current of 10mA at room temperature.
 A) 4Ω B) 258Ω C) 2.6Ω D) 0.4Ω
119. The guaranteed accuracy of a 0-10A ammeter is 1% of full scale deflection. Estimate the limiting error for a reading of 2.5A:
 A) 1% B) 4% C) 2% D) Infinite
120. The form factor of a half-wave rectifier is:
 A) $\pi/2$ B) $\pi/\sqrt{2}$ C) $\sqrt{2}$ D) $\pi/(2\sqrt{2})$
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